

Design and Evaluation of a Low-Cost Educational Network Segmentation Framework Guided by Zero Trust Principles

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Schools and universities rely more and more on digital networks these days—for everything from teaching and learning to managing admin tasks, running research projects, hosting online platforms, and connecting everyone through Wi-Fi. This means campus networks have grown pretty complex, linking all kinds of users, devices, servers, apps, and even Internet of Things gadgets all in one system. While this makes things run smoother and faster, it also opens the door to more cybersecurity risks from both inside and outside the campus.

In many places, network access is pretty open and flexible by design. Students, teachers, staff, visitors, and tech folks all need different levels of access to get their work done. But at the same time, these networks hold sensitive stuff like grades, financial info, admin databases, learning systems, file storage, and network controls. If the network isn't carefully divided and locked down, one hacked device could be the weak link that lets attackers sneak into other vital parts of the system.

Traditional network security often relies on guarding the perimeter with firewalls and access controls, trusting anything inside to be safe. But these days, that kind of thinking just doesn't cut it. Attackers can sneak in through phishing, weak passwords, malware, or unsecured Wi-Fi. Once inside, they can poke around, escalate their privileges, and move sideways to reach sensitive data.

That's where Zero Trust Architecture steps in. It flips the old idea on its head by saying no one and nothing inside the network should be trusted automatically. Every user, device, and app has to prove they belong, access is kept tight, and activity is watched all the time. This way of thinking fits perfectly with today's world where users and devices are everywhere, always changing.

One key way to make Zero Trust work is through micro-segmentation. This breaks the network into tiny chunks and controls who can talk to what, based on clear rules. Instead of letting everything inside talk freely, it locks down communication to only what's needed. This helps stop attackers from hopping around, limits damage if something's compromised, and gives better insight into what's happening inside the network.

People are talking more and more about Zero Trust and micro-segmentation in cybersecurity circles, but most of the real-world setups we see are built for big companies, cloud services, or fancy data centers. These places usually have the budget for high-end gear and experts to manage it all. That leaves smaller schools and colleges stuck—they want better network security but don't have the money or tech teams to pull off these complex solutions.

That's why this study is all about creating a simple, budget-friendly Zero Trust micro-segmentation approach tailored for educational institutions. The idea is to use everyday tech like VLANs, firewall rules, and intrusion detection systems—things schools can actually access. We'll build this framework in a virtual network and test it by comparing three setups: a flat network with no segmentation, a basic VLAN-segmented network, and the new Zero Trust micro-segmented network we're proposing.

1.2 Problem Statement

A lot of schools and colleges still use simple network setups or just basic VLANs to separate their devices. In these kinds of networks, there's not much control over how users, devices, and services talk to each other inside. That leaves the door open for security problems—like someone getting in without permission, scanning the network, spreading malware, or moving around once they've hacked one device or account.

When a network is flat or only loosely divided, it makes the whole institution more vulnerable. For instance, if a student's laptop or a guest's device gets compromised, they might be able to explore the internal network, find open ports, and try to break into servers or admin systems. Even with VLANs set up, if there aren't strict firewall rules or constant monitoring between them, the protection isn't strong enough.

There are commercial Zero Trust and micro-segmentation tools out there that can help fix these issues, but they tend to be pricey, complicated, and built mostly for big companies. Smaller schools often don't have the budget or tech know-how to use those solutions. That's why there's a real need for a straightforward, affordable, and measurable approach that brings Zero Trust micro-segmentation to life using the network tools schools already have—like VLANs, firewalls, and IDS.

So, the challenge here is clear: many educational networks don't have a practical, low-cost way to segment their internal systems that stops attackers from moving sideways, blocks unauthorized communication between different parts of the network, and helps spot suspicious activity—all using common tools they can actually manage.

1.3 Research Gap

Earlier research and official cybersecurity advice have stressed how crucial Zero Trust Architecture, least privilege access, constant monitoring, and micro-segmentation really are. People have looked into Zero Trust across cloud setups, data centers, software-defined networks, multi-cloud environments, and big enterprise systems. Some studies even suggest advanced micro-segmentation methods using stuff like SDN, VXLAN, AI, or commercial Zero Trust tools.

But when it comes to practical, budget-friendly Zero Trust micro-segmentation frameworks tailored for school networks, there's still not much out there. Most solutions are either too theoretical, expensive, or just too complicated. They often don't fit the needs of small to medium schools, which usually rely on traditional networks and open-source or common tools.

On top of that, few studies actually compare a flat network, traditional VLAN segmentation, and a Zero Trust micro-segmentation setup side-by-side, using real security and performance data. This leaves a clear gap—and a chance to build and test a down-to-earth framework that schools can set up with VLANs, firewall rules, and IDS in a simple virtual lab.

1.4 Aim of the Study

This research sets out to create and test an affordable Zero Trust Micro-Segmentation system to protect the networks of educational institutions. It focuses on using VLANs, firewall rules, and an intrusion detection system to keep things safe.

1.5 Research Objectives

1. Take a close look at the security weaknesses in flat and traditionally segmented networks used by schools and colleges.
2. Come up with a realistic Zero Trust Micro-Segmentation setup that fits well with the way educational networks work.
3. Bring that setup to life using VLANs, firewall rules, and intrusion detection systems in a virtual network environment.
4. Put the new framework side-by-side with flat networks and old-school VLAN setups to see how they stack up.
5. Check how well the new system works by measuring security and performance with clear numbers.

6. Create a straightforward security policy chart that spells out what kind of communication is allowed or blocked between different parts of the network.

7. Share practical advice for schools and colleges on how to boost their internal network security without spending a fortune.

1.6 Research Questions

1. How can we create and test a budget-friendly Zero Trust Micro-Segmentation setup that cuts down on sideways hacking and blocks unauthorized access in school networks?

2. What are the biggest security weaknesses in flat or old-school segmented networks used by schools?

3. How can VLANs, firewall rules, and intrusion detection systems work together to bring Zero Trust Micro-Segmentation to the network level?

4. How much can this new setup actually stop unauthorized chats between different parts of the network?

5. What changes happen with this framework when it comes to hackers moving around, services being exposed, open ports, blocked connections, and alerts from the IDS?

6. Does this framework slow things down? What's its impact on delay, data flow speed, and lost packets?

7. How can a clear policy guide help make Zero Trust Micro-Segmentation a reality in school networks?

1.7 Significance of the Study

On the academic side, this research adds a fresh angle to network security by bringing Zero Trust and micro-segmentation into the world of school networks. While a lot of studies zoom in on cloud setups or big enterprise systems, this one looks at something more hands-on—how schools can use simple, easy-to-get tech to build a stronger network. It also offers a side-by-side look at three types of network setups: a flat network, the usual VLAN segmentation, and the Zero Trust micro-segmentation approach.

From a real-world standpoint, this study could be a game-changer for schools trying to beef up their network security without shelling out big bucks for fancy software. The framework it suggests helps schools split users, devices, and services into safe zones, cutting down on unauthorized access and making it easier to spot sketchy activity through monitoring tools.

This is a big deal, especially for schools that don't have a lot of money or tech experts on hand.

In the end, the study aims to deliver an easy-to-follow, reusable plan that lays out how to set up network zones, firewall rules, where to put monitoring tools, and a security policy guide. This kind of roadmap could really help network admins build safer school networks using affordable, widely available tech.

1.8 Scope of the Study

This study takes a close look at how to bring Zero Trust Micro-Segmentation to the network layer in schools and universities. The plan is to build the setup using VLANs, firewall rules, and intrusion detection systems. Instead of testing this out on a real, live network, everything will be done in a virtual lab to keep things safe and controlled.

The research will compare three different setups: a flat network with no segmentation, a standard VLAN-based segmentation, and the new Zero Trust Micro-Segmentation approach they're proposing.

It's important to note that this study won't cover a full-blown enterprise Zero Trust system. It won't dive into things like identity management, multi-factor authentication, endpoint detection, cloud-native Zero Trust, or fully software-defined perimeters. The focus is more practical—applying Zero Trust ideas right at the network layer using affordable and straightforward tech.

The different parts of the network being looked at include:

- Student network
- Staff network
- Administration network
- Server network
- Guest network
- IoT or CCTV network
- Network management zone
- Internet access zone

1.9 Definition of Key Terms

Zero Trust Architecture: A cybersecurity approach based on the principle that no user, device, system, or network segment should be trusted by default. Access should be continuously verified and limited according to need.

Micro-Segmentation: A network security technique that divides the network into smaller security zones and controls traffic between them using strict access policies.

VLAN: A Virtual Local Area Network used to logically separate devices within the same physical network infrastructure.

Firewall Policies: Rules configured on a firewall to allow or deny traffic based on source, destination, port, protocol, and security requirements.

Intrusion Detection System: A security tool that monitors network traffic and generates alerts when suspicious or malicious activity is detected.

Lateral Movement: The technique used by attackers to move from one compromised system to other systems within the same network.

Flat Network: A network design where devices have broad internal access with minimal segmentation or access control.

Least Privilege: A security principle that gives users, devices, and services only the minimum level of access required to perform their tasks.

Default Deny: A security rule approach where all traffic is blocked by default unless it is explicitly allowed.

1.10 Chapter Summary

This chapter laid out the background, the problem at hand, and the gap that needs filling. It also shared the study's aim, objectives, research questions, importance, scope, and key terms. It pointed out that networks in educational institutions are facing more internal security threats because of their complexity, the need for open access, and poor segmentation. The chapter emphasized the need for a practical, budget-friendly Zero Trust Micro-Segmentation framework to help cut down on sideways movement within the network and stop unauthorized access, using tools like VLANs, firewall rules, and intrusion detection systems. The next chapter will dive into existing research on Zero Trust Architecture, micro-segmentation, VLAN-based segmentation, firewall policies, intrusion detection systems, and related studies.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter takes a close look at the key ideas, standards, and past research tied to the study at hand. It dives into topics like network security in schools, the usual ways networks are divided up, Zero Trust Architecture, micro-segmentation, VLAN-based segmentation, firewall rules, and intrusion detection systems. It also highlights some important studies that focus on Zero Trust micro-segmentation and real-world network security setups.

The goal here is to lay down a solid mix of theory and hands-on knowledge that supports this research. It also points out the gap this study wants to fill. From what's been reviewed, it's clear that Zero Trust and micro-segmentation are gaining ground as smart ways to limit sideways attacks and boost security inside networks. But, most current solutions are built for big companies, cloud systems, data centers, or software-defined networks. That leaves a real need for a straightforward, budget-friendly framework that fits the unique needs of educational institutions.

2.2 Educational Institution Network Security

Schools and universities lean heavily on their network systems to keep everything running—from teaching and learning to research, admin tasks, communication, and just getting online. Their networks have a mix of users: students, teachers, staff, guests, and tech teams. Plus, they often run things like learning platforms, file storage, databases, security cameras, Wi-Fi spots, and all kinds of smart devices.

Because of this mix, these networks aren't like your typical company setups. They need to be open and flexible, especially for students, visitors, researchers, and folks using wireless connections. But at the same time, they hold really sensitive info—student records, financial data, academic tools, staff details, and admin systems. Balancing openness with protecting this sensitive stuff is tough.

One big worry is what's called lateral movement. If a hacker gets into one device—say on the student or guest Wi-Fi—they might be able to move around inside the network if it's not properly segmented. They could scan for other devices, find weak spots, and try to

break into important servers or admin systems. That's why controlling internal access is so important to keeping educational networks safe.

2.3 Traditional Network Security and Its Limitations

Traditional network security usually focuses on keeping threats out by building a strong perimeter. This means putting firewalls, access controls, and intrusion detectors at the edges of the network to watch over traffic coming in and going out. It's like locking the front door and hoping that keeps everyone safe inside.

But these days, that's not enough. Attackers can sneak in from within—through hacked devices, stolen passwords, phishing scams, infected USB drives, weak Wi-Fi, or vulnerable internal systems. Once they're inside, the perimeter defenses don't do much to stop them.

Another problem with the old way is assuming that everything inside the network is trustworthy. That's risky, especially in places like schools where many users and devices have different needs and access levels. If the network isn't divided up carefully, one hacked device could open the door to a lot more. That's why modern security focuses on controlling what each user and device can do inside the network—giving access only where it's truly needed based on who they are, what device they're using, and what services they require

2.4 Zero Trust Architecture

Zero Trust Architecture is a fresh way of thinking about cybersecurity. It's built on the idea that no user, device, or part of the network should be trusted just because it's inside the system. Instead of assuming everything within the network is safe, Zero Trust means constantly checking and controlling who and what can access resources.

According to the National Institute of Standards and Technology, Zero Trust shifts the focus from guarding the network's outer edges to protecting the people, devices, and data themselves. It's about designing and defending enterprise systems using these core principles (Rose et al., 2020).

At its heart, Zero Trust means never trusting by default, always verifying, giving users only the access they absolutely need, assuming breaches can happen, watching and logging activities, limiting access strictly, and regularly reviewing access permissions.

For schools and universities, this approach is especially helpful. It cuts down the chances of someone inside the campus network getting too much access just because they're "inside." Every connection and communication needs a good reason and must follow clear rules.

That said, fully adopting Zero Trust can be tricky. It often calls for tools like identity management, multi-factor authentication, checking device health, nonstop monitoring, au-

tomation, and tying together different security systems. So, this study zooms in on applying Zero Trust ideas at the network level using micro-segmentation, VLANs, firewall rules, and intrusion detection systems to make it more manageable.

2.5 Micro-Segmentation

Micro-segmentation is a security method that breaks a network into smaller chunks or zones. Each zone holds certain users, devices, apps, or services. Instead of letting everything communicate freely, security rules control how these zones talk to each other.

The big idea behind micro-segmentation is to stop attackers from roaming around easily. If someone hacks into one device or zone, this setup makes it much harder for them to get into other parts of the network. It helps keep damage in check and protects sensitive stuff.

CISA calls micro-segmentation a key step in moving toward Zero Trust security. Instead of just focusing on keeping threats out at the network's edge, it's about creating tighter trust zones inside. It makes sure communication inside the network is limited, watched closely, and follows strict policies.

You can build micro-segmentation using lots of tech tools—VLANs, firewall rules, software-defined networking, host firewalls, cloud security groups, identity-based controls, and more. For smaller schools or organizations, using VLANs combined with firewall settings and intrusion detection can be a budget-friendly way to get many of the benefits while sticking to Zero Trust ideas.

2.6 VLAN-Based Network Segmentation

A Virtual Local Area Network, or VLAN, is a handy way to keep devices separated even when they're all plugged into the same physical network. It lets network admins group people and gadgets by what they do, their department, role, or security needs.

In schools, for example, VLANs can keep students, staff, admin, servers, guests, IoT devices, CCTV cameras, and network gear all in their own lanes. This setup helps keep things organized and cuts down on unnecessary traffic blasting across the network. Plus, it gives a simple layer of separation between different user groups.

But VLANs by themselves aren't a security silver bullet. If traffic is allowed to flow freely between VLANs without tight controls, someone on one VLAN might still get into sensitive stuff on another. That's why VLANs should be paired with firewall rules. The firewall can then control who talks to who, based on things like where the traffic's coming from, where it's heading, what protocol or port it's using, and what service is involved. This way, only the

necessary communication is allowed, sticking to the principle of least privilege and keeping the network zones safer.

2.7 Firewall Policies and Access Control

Firewall policies play a big role in controlling how different parts of a network talk to each other. A firewall rule can either let traffic through or block it, based on things like where it's coming from, where it's going, the type of protocol, port, app, or even the direction of the traffic.

In a Zero Trust micro-segmentation setup, the best practice is to block everything by default. In other words, no communication happens between segments unless there's a solid reason to allow it. Only the services that are absolutely needed get through.

For instance, the Guest VLAN should only be able to browse the internet. The Student VLAN gets access to the internet plus the learning platform. The Administration VLAN can reach the admin servers, and the IoT/CCTV VLAN should only talk with its management or recording server. This kind of policy keeps the network safer by cutting down the number of places attackers can get in and stopping unnecessary access inside the network.

2.8 Intrusion Detection Systems in Segmented Networks

An Intrusion Detection System keeps an eye on network traffic, looking out for anything fishy or harmful. Tools like Snort and Suricata can spot things like scanning attempts, known attack patterns, unusual traffic flows, and breaches of set rules.

In a network broken into segments, IDS gives you a clear view of what's happening between those sections. That's key because micro-segmentation isn't just about blocking unwanted traffic—it's also about catching sneaky tries to break the rules.

For instance, if a device in the Student VLAN tries to scan the Server VLAN, the firewall might stop the traffic, but the IDS will raise an alert. These alerts help network admins get a handle on what kind of suspicious activity is going on inside and how often. IDS tools are especially handy in studies like this because they provide clear, measurable results.

2.9 Review of Related Studies

2.9.1 NIST Zero Trust Architecture

NIST SP 800-207 is a key resource when it comes to Zero Trust Architecture. It shifts the focus away from traditional, fixed network boundaries and instead puts the spotlight on

protecting users, assets, and resources wherever they are. This document lays down the core ideas behind Zero Trust, which is why it's so important for this study. Building on that, the research zooms in on a specific piece of Zero Trust—network-layer micro-segmentation. Rather than trying to roll out the entire Zero Trust model, it applies these principles to internal network traffic using tools like VLANs, firewall rules, and intrusion detection systems.

2.9.2 CISA Microsegmentation in Zero Trust

CISA's guidance on Microsegmentation in Zero Trust brings fresh, official backing to the idea that micro-segmentation plays a crucial role in making Zero Trust work. It stresses the importance of careful planning, setting clear trust boundaries, and managing how devices communicate inside the network. This fits perfectly with the current study, which aims to limit uncontrolled communication inside the network. It also agrees that micro-segmentation needs to be thoughtfully designed around network assets, communication patterns, and security rules.

2.9.3 NSA/CISA Zero Trust Network and Environment Pillar

The NSA and CISA's advice on the Network and Environment Pillar focuses on securing network traffic, boosting visibility, and cutting down on lateral movement (where attackers move sideways inside a network). This is directly relevant to the research since the proposed framework is all about managing internal traffic between different zones of an educational institution's network.

2.9.4 Moving Toward a Zero-Trust Micro-Segmentation Security Approach

Basta and colleagues came up with a way to measure how well Zero Trust micro-segmentation works. They used graph-based connectivity and attack graph analysis to see how exposed and strong a network really is. Their findings showed that micro-segmentation can greatly cut down exposure and make networks tougher against attacks.

This is really helpful for our research because it backs up using clear, measurable ways to judge micro-segmentation's impact. But their work leans more toward analysis and doesn't offer a simple, budget-friendly setup tailored for schools. Our study takes it further by actually building and testing a practical system using VLANs, firewall rules, and intrusion detection systems.

2.9.5 Micro-Segmentation Using VLAN-VXLAN Mapping Technology

Li and team introduced a micro-segmentation method that links VLAN and VXLAN. Their work is relevant because it shows you can do network segmentation without breaking the

bank. Plus, they proved you can use standard network protocols without shaking up existing services too much.

That said, their focus is mostly on data centers and the VLAN-VXLAN setup. Our research takes a different path by zeroing in on school networks and putting together a low-cost, hands-on framework using VLANs, firewall policies, and IDS.

2.9.6 ZT-SDN: Bringing Zero Trust to Software-Defined Networks

ZT-SDN is a newer approach that takes Zero Trust ideas and applies them to software-defined networks. It zeroes in on giving users only the access they really need and cutting down on sideways movement across the network, using smart tools like SDN and machine learning.

This area is pretty hot in research right now, with Zero Trust and micro-segmentation leading the way in network security. But the catch is, using SDN and machine learning often means you need some pretty advanced skills and setup. That's why this study offers a simpler, more hands-on method for schools, using tools they probably already have on hand.

2.9.7 Microsegmented Cloud Network Architecture with Open-Source Tools

Arora and Hastings took on the challenge of building a cloud network that uses micro-segmentation and Zero Trust, relying solely on open-source tools. Their work proves you don't have to lean fully on pricey commercial security software to get solid protection.

Building on that practical vibe, this research shifts focus to school networks instead of cloud setups. It dives into using VLANs for segmentation, setting firewall rules, and keeping an eye on things with IDS monitoring.

2.9.8 Comparing Snort and Suricata

Lots of studies have looked at how Snort and Suricata stack up as open-source intrusion detection systems. This matters because the framework here includes IDS as a key piece. Both Snort and Suricata can watch network traffic, spot scanning attempts, and send out alerts during tests.

Depending on the lab setup chosen, this research might go with Snort or Suricata. Either way, the IDS findings will be a big part of measuring how well the proposed framework works.

2.10 Comparison of Previous Studies

The following table summarizes the relationship between previous studies and the current research.

Study / Source	Focus	Environment	Technology	Limitation	Relevance to Current Study
NIST SP 800-207	Zero Trust Architecture	Enterprise environments	Policy-based Zero Trust model	General framework, not a specific implementation	Provides theoretical foundation
CISA Microsegmentation in Zero Trust	Micro-segmentation planning	Enterprise networks	Trust boundaries and segmentation planning	Guidance document, not an experimental study	Supports micro-segmentation concept
NSA/CISA Network and Environment Pillar	Zero Trust network security	Enterprise networks	Network visibility and segmentation	General guidance	Supports internal traffic control
Basta et al.	Evaluation of micro-segmentation	Enterprise networks	Connectivity and attack graphs	Analytical evaluation, not educational network implementation	Supports quantitative evaluation
Li et al.	VLAN-VXLAN micro-segmentation	Data centers	VLAN/VXLAN mapping	Focused on data center traffic	Supports network-based segmentation
ZT-SDN	Zero Trust in SDN	SDN environment	SDN and machine learning	Requires advanced infrastructure and programming	Shows modern direction but differs from proposed low-cost approach
Arora and Hastings	Open-source microsegmented architecture	Cloud / multi-cloud	Open-source tools	Cloud-focused	Supports low-cost and open-source implementation
Snort/Suricata studies	IDS comparison	Network security labs	IDS tools	Tool-focused, not full segmentation framework	Supports IDS selection and monitoring

2.11 Research Gap Summary

Looking over the existing research, it's clear that Zero Trust and micro-segmentation play a big role in beefing up internal network security and cutting down on lateral movement. Organizations like NIST, CISA, and NSA/CISA all push for things like strict access control, least privilege, better network visibility, and solid segmentation.

But there are still some holes. For starters, most studies zoom in on cloud setups, data centers, SDN, or big enterprise networks. Then, a lot of the solutions rely on fancy tech, commercial tools, or programming-heavy automation—things that smaller schools and colleges might struggle to use. Also, there aren't many practical, budget-friendly frameworks out there that mix VLANs, firewall rules, and IDS specifically for educational networks. Finally, there's a lack of real-world comparisons between flat networks, traditional VLANs, and Zero Trust micro-segmentation using clear-cut security and performance measures.

That's where this study steps in. It aims to build and test an affordable, network-layer Zero Trust Micro-Segmentation framework tailored for educational institutions. The plan is to use easy-to-get technologies and back it up with solid numbers—looking at things like how many hosts are reachable, open ports, blocked unauthorized access, success rates of lateral movement, IDS alerts, latency, throughput, and packet loss.

2.12 Chapter Summary

This chapter took a close look at the research around network security in schools and colleges. It covered the usual limits of traditional security, what Zero Trust Architecture brings to the table, and ideas like micro-segmentation and VLANs. It also dug into firewall rules and intrusion detection systems. Some studies and official advice tied to this research were reviewed too. The takeaway? Zero Trust micro-segmentation helps stop attackers from moving around inside a network and boosts security overall. But most solutions out there focus on big businesses, cloud setups, or data centers—not really on education networks. That shows there's a real need for a simple, budget-friendly approach designed specifically for schools. Up next, the following chapter will lay out the research methods, the proposed framework, the testing setup, how success will be measured, and what results are expected.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

In this chapter, I'll walk you through the research methods I'll use to create and test a low-cost Zero Trust Micro-Segmentation framework tailored for school networks. I'll cover everything—from how the study is set up, to the framework itself, the test environment, different network setups, security zones, policy rules, tools, ways to gather data, how I'll measure success, analyze results, and even the ethical stuff, limitations, and the timeline.

The main goal here is to lay out a clear, step-by-step plan for how the research will happen. Since I'm aiming to build a practical security framework and see how well it works, the approach mixes design with hands-on testing. I'll run experiments in a virtual network lab, putting three different setups head-to-head: a simple flat network, a network segmented by traditional VLANs, and the new Zero Trust Micro-Segmented network I'm proposing. This way, it's easier to compare and understand the benefits of each.

3.2 Research Design

This study takes a hands-on, numbers-driven approach grounded in Design Science Research. It's practical because it tackles a real network security challenge faced by educational institutions. Instead of just talking about Zero Trust and micro-segmentation in theory, it offers a concrete framework that anyone can put to work using readily available technology.

The study leans on quantitative methods since it measures how well the framework performs through clear metrics like how many hosts can be reached, open ports, blocked unauthorized connections, IDS alerts, success rate of lateral movement, latency, throughput, and packet loss.

Following the Design Science Research path, the study builds and tests something tangible—the low-cost Zero Trust Micro-Segmentation framework. This framework isn't just designed and implemented; it's also rigorously tested and assessed based on security effectiveness and network performance.

3.3 Research Methodology Steps

Step 1: Literature Review

I'll dive into past studies, standards, and technical papers about Zero Trust Architecture, micro-segmentation, VLANs, firewall rules, IDS, and how networks are secured in schools and universities.

Step 2: Requirement Analysis

Next, I'll figure out what a typical school network really needs to stay safe—like keeping students, staff, admins, servers, guests, IoT gadgets, and management devices separate.

Step 3: Framework Design

Then, I'll sketch out the Zero Trust Micro-Segmentation framework. This will cover network zones, how VLANs are set up, firewall rules, where to put the IDS, and a clear security policy guide.

Step 4: Virtual Lab Implementation

After that, I'll bring the framework to life in a virtual lab using tools like GNS3 or EVE-NG, pfSense, Snort or Suricata, Kali Linux, Nmap, Wireshark, and iPerf.

Step 5: Experimental Testing

I'll run tests on three setups: a flat network, traditional VLAN segmentation, and the new Zero Trust Micro-Segmentation framework to see how they stack up.

Step 6: Data Collection and Analysis

I'll gather data from firewall logs, IDS alerts, Nmap scans, Wireshark captures, ping tests, and iPerf results to get a solid picture of how each setup performs.

Step 7: Evaluation and Discussion

Then, I'll compare the results to judge if the new framework really boosts security without slowing things down too much.

Step 8: Conclusion and Recommendations

Finally, I'll wrap up with key takeaways, practical advice, any research limitations, and ideas for what to explore next.

3.4 Proposed Framework Overview

This framework is all about bringing Zero Trust Micro-Segmentation to school networks without breaking the bank. It uses VLANs to keep groups separate, firewall rules to manage who talks to whom, and an IDS to keep an eye out for any shady activity inside the network.

It's not trying to build a full-on enterprise Zero Trust system. Instead, it focuses on practical, affordable tools that schools can actually use to put Zero Trust ideas into action at the network level.

Some key ideas here:

- Deny everything by default
- Only give access that's absolutely needed
- Use clear rules to allow communication
- Separate users and devices into segments
- Control how VLANs interact
- Keep an eye on internal traffic
- Make it harder for attacks to spread sideways
- Limit the number of open services
- Log and analyze anything suspicious

3.5 Proposed Network Zones

The plan is to split the school's network into different security zones. Each zone will group users, devices, or services that share similar access needs, helping keep things tidy and secure.

Zone	Description
Student Zone	Student devices connected through wired or wireless access. Students should have limited access to the internet and selected academic services.
Staff Zone	Academic and teaching staff devices requiring access to academic platforms, selected internal services, and the internet.
Administration Zone	Administrative staff devices requiring controlled access to administrative systems, databases, and internal services.
Server Zone	Institutional servers such as learning management systems, file servers, database servers, and application servers.

Zone	Description
Guest Zone	Visitors and guest users who should only have internet access and no access to internal institutional systems.
IoT/CCTV Zone	Cameras, IoT devices, printers, or other smart devices that should communicate only with required management or recording servers.
Management Zone	Network administration devices used to manage switches, firewalls, servers, and IDS systems.
Internet Zone	External internet access and public network communication.

3.6 Security Policy Matrix

The security policy matrix lays out which zones can talk to each other and which can't. It's built on a default deny rule, meaning everything is blocked by default unless there's a clear permission to allow it.

Source Zone	Destination Zone	Access Decision	Allowed Services	Security Justification
Guest Zone	Internet	Allow	HTTP, HTTPS, DNS	Guests need internet access only
Guest Zone	Internal Zones	Deny	None	Prevent access to internal resources
Student Zone	Internet	Allow	HTTP, HTTPS, DNS	Students need internet access
Student Zone	Learning Server	Allow	HTTP, HTTPS	Students need access to academic platform
Student Zone	Administration Zone	Deny	None	Prevent access to sensitive systems
Student Zone	Management Zone	Deny	None	Prevent unauthorized network management access
Staff Zone	Internet	Allow	HTTP, HTTPS, DNS	Staff need internet access
Staff Zone	Learning Server	Allow	HTTP, HTTPS	Staff need access to academic services
Staff Zone	Administration Zone	Deny or Restricted	Based on role	Limit unnecessary access
Administration Zone	Administrative Server	Allow	Required ports only	Administrative work requirement
IoT/CCTV Zone	Recording Server	Allow	Required service ports	CCTV/IoT operational need
IoT/CCTV Zone	Other Internal Zones	Deny	None	Limit IoT lateral movement risk
Management Zone	Network Devices	Allow	SSH, HTTPS, SNMP if required	Network administration

Source Zone	Destination Zone	Access Decision	Allowed Services	Security Justification
All Zones	All Unrequired Destinations	Deny	None	Default-deny and least privilege

3.7 Experimental Network Scenarios

3.7.1 Scenario One: Flat Network

Here, all devices connect in a simple, flat network with almost no internal restrictions. This setup reflects a weak internal security model where devices can easily talk to one another. The goal is to establish a baseline—see how risky it is when there's little to no segmentation. We'll check things like open ports, how many devices can be reached, and how easily an attacker could move around. In this setup, a hacker might scan lots of devices, find open services, and try to get into sensitive servers or equipment.

3.7.2 Scenario Two: Traditional VLAN Segmentation

This time, the network is split into VLANs—think Student VLAN, Staff VLAN, Admin VLAN, Server VLAN, Guest VLAN, IoT/CCTV VLAN, and Management VLAN. There will be some basic routing or limited access controls between them. This reflects a common way networks are segmented today. The idea is to see if just using VLANs cuts down attack risks and to compare this with the Zero Trust approach later. VLANs can block some direct communication, but if firewalls aren't tight, devices might still talk more than they should.

3.7.3 Scenario Three: Proposed Zero Trust Micro-Segmentation Framework

In this setup, the network is carefully sliced into security zones using VLANs, with strict firewall rules controlling traffic based on a detailed security policy matrix. IDS tools will watch for suspicious activity like scanning or unauthorized attempts. The goal here is to block everything by default, only allowing what's absolutely necessary. This should cut down unauthorized access, lateral movement, and make it easier to spot suspicious behavior inside the network. Compared to the first two scenarios, this approach should significantly reduce the number of reachable hosts, open ports, and chances for attackers to move around unnoticed.

3.8 Experimental Environment

The research will take place in a controlled virtual lab setup. Using a virtual lab works well for this study since it lets us test security scenarios safely, without risking any real institutional systems.

Tool	Purpose
GNS3 or EVE-NG	Design and simulate the network topology
pfSense	Firewall and router to enforce inter-VLAN firewall policies
Snort or Suricata	IDS to monitor network traffic and generate alerts
Kali Linux	Testing machine to perform scanning and controlled attack simulations
Ubuntu Server or Windows Server	Internal servers such as learning platform, file server, or administrative server
Wireshark	Capture and analyze network traffic
Nmap	Scan hosts and open ports
iPerf	Measure throughput
Ping and Traceroute	Test reachability and latency

3.9 Proposed Network Topology

This network design is meant to show a basic layout for a school or college system. At the center, there's a firewall or router keeping things secure. Around it, there are several VLANs set up to separate different groups of users and devices. Inside, you'll find a few servers handling various tasks, plus a dedicated management network to keep control smooth. There's also a machine acting as a test attacker to check for vulnerabilities, an IDS to keep an eye on any suspicious activity, and of course, access to the internet or a simulated version of it.

VLAN ID	Zone Name	Description
VLAN 10	Student Zone	Student devices
VLAN 20	Staff Zone	Academic staff devices
VLAN 30	Administration Zone	Administrative users
VLAN 40	Server Zone	Internal servers
VLAN 50	Guest Zone	Guest internet users
VLAN 60	IoT/CCTV Zone	Cameras and IoT devices
VLAN 70	Management Zone	Network management
VLAN 80	IDS/Monitoring Zone	IDS and logging tools

3.10 Data Collection Methods

Data will be collected from several sources during the experimental testing.

Data Source	Collected Data
Firewall Logs	Allowed and blocked connections between network zones
IDS Alerts	Suspicious activity such as scanning, unauthorized access attempts, and abnormal traffic
Nmap Scan Results	Reachable hosts and open ports in each scenario
Wireshark Captures	Network traffic and communication behavior between zones
Ping Results	Reachability and latency
iPerf Results	Throughput between selected network segments
Test Records	Allowed and denied access attempts based on the policy matrix

3.11 Evaluation Metrics

3.11.1 Security Metrics

Metric	Description
Number of Reachable Hosts	Measures how many devices or servers can be reached from a given network zone
Number of Open Ports	Measures how many service ports are exposed during scanning
Blocked Unauthorized Connections	Measures how many unauthorized attempts are blocked by firewall policies
Lateral Movement Success Rate	Successful unauthorized access attempts divided by total unauthorized access attempts, multiplied by 100
Attack Surface Reduction	Reduction in reachable services and exposed ports after applying the proposed framework
IDS Alert Count	Number of alerts generated by the IDS during scanning or unauthorized access attempts
Policy Violation Attempts	Attempts that violate the defined security policy matrix

3.11.2 Performance Metrics

Metric	Description
Latency	Measured using ping between selected zones

Metric	Description
Throughput	Measured using iPerf between selected zones
Packet Loss	Measured during communication tests
Performance Overhead	Compares performance before and after applying strict segmentation and firewall rules

3.12 Testing Procedures

1. Connectivity Testing: We'll use ping and traceroute to see if different zones can talk to each other.
2. Port Scanning: Nmap will help us check the open ports on selected hosts across zones.
3. Unauthorized Access Testing: We'll try, in a controlled way, to access services that should be off-limits from unauthorized zones.
4. Lateral Movement Testing: Starting from a less trusted zone like Student or Guest, the attacker machine will try to move into other internal zones.
5. IDS Monitoring: We'll keep an eye on the IDS logs to catch any alerts triggered during scanning or access attempts.
6. Performance Testing: We'll measure things like latency, throughput, and packet loss to see how the new setup affects performance.
7. Result Recording: All the findings will be neatly recorded in tables to compare across the three scenarios.

3.13 Data Analysis

We'll analyze the collected data using straightforward stats and side-by-side comparisons. The focus will be on how the three scenarios stack up when it comes to security and performance. We'll show the findings through tables, charts, and calculations of percentage reductions.

The framework will be considered a success if it cuts down on unauthorized access, open ports, chances for lateral movement, and exposed services—all while keeping network performance within a reasonable range.

Here's what we'll look at:

- How many hosts can be reached in each scenario

- The number of open ports found by Nmap
- How many unauthorized connections get blocked
- The success rate of lateral movement attempts
- How much the attack surface shrinks
- IDS alerts across the different setups
- Changes in latency and throughput before and after applying the framework

3.14 Expected Results

1. Less unauthorized access across network zones
2. Fewer chances for attackers to move laterally
3. A drop in reachable hosts from untrusted areas
4. Fewer exposed services and open ports
5. More unauthorized connections getting blocked
6. Better visibility into suspicious internal traffic via IDS alerts
7. A clear, reusable policy matrix that educational institutions can adopt
8. Only a small hit on network performance in terms of latency and throughput

Ultimately, we hope to prove that a low-cost Zero Trust Micro-Segmentation setup can boost security inside educational networks without the need for pricey commercial solutions

3.15 Ethical Considerations

This research will take place entirely within a controlled virtual lab. No real networks, user data, or live systems will be involved. All tests will run strictly on authorized virtual machines and simulated network setups. The study won't touch any real systems or outside networks. Tools like Nmap and Kali Linux will be used only for learning and research within this safe environment. The whole process will stick to responsible and ethical cybersecurity practices.

3.16 Research Limitations

1. The work happens in a virtual lab, not an actual campus network.

2. The virtual setup might miss some of the real-world complexity found in big educational networks.

3. This study focuses on network-level Zero Trust Micro-Segmentation, not the full identity-based Zero Trust approach.

4. It doesn't cover things like multi-factor authentication, endpoint detection, device health checks, or full commercial Zero Trust solutions.

5. Results could change depending on network size, traffic levels, firewall settings, and the IDS tools used.

6. Testing uses controlled scenarios, not real user traffic.

3.17 Research Timeline

The research is expected to be completed within ten months.

Month	Activity
Month 1	Collect and review literature related to Zero Trust, micro-segmentation, VLANs, firewall policies, and IDS
Month 2	Finalize research problem, objectives, questions, and methodology
Month 3	Design the proposed framework, network zones, and security policy matrix
Month 4	Build the virtual lab environment using GNS3 or EVE-NG
Month 5	Implement the flat network scenario and collect baseline results
Month 6	Implement the traditional VLAN segmentation scenario and collect results
Month 7	Implement the proposed Zero Trust Micro-Segmentation scenario
Month 8	Run security and performance tests across the three scenarios
Month 9	Analyze the collected data and prepare results and discussion
Month 10	Complete writing, review, editing, and final submission

3.18 Chapter Summary

In this chapter, the research approach was laid out clearly. The study will follow an applied quantitative design rooted in Design Science Research. The plan is to build and test the framework in a virtual lab using three different setups: a flat network, traditional VLAN segmentation, and Zero Trust Micro-Segmentation. The chapter also went over the network zones proposed, the policy matrix, tools for experimentation, how data will be gathered, the metrics for evaluation, testing steps, how the data will be analyzed, along with ethical concerns, study limitations, and the overall research schedule.

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